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Lecture No.	*Topic Covered*	*Page No.*	*Total Pages*
1	Introduction Least Squares Estimation in Linear Regression Models	1-5	5
2	Statistical Inference in Linear Regression, Prediction of New Observations	6-10	5
3	Model Adequacy Checking	11-14	4
4	Variable Selection Methods in Regression	15-18	4
5	Generalized and Weighted Least Squares	19-21	3
6	Regression Models for General Time Series Data	22-25	4
7	Exponential Smoothing	26-28	3
8	First order and Second order	29-31	3

References/Text Books

1	Introduction To Time Series Analysis And Forecasting, 2nd Edition, Wiley Series In Probability And Statistics, By Douglas C. Montgomery, Cheryl L. Jen(2015)
2	Master Time Series Data Processing, Visualization, And Modeling Using Python Dr. Avishek Pal Dr. Pks Prakash (2017)
3	Ruey S. Tsay "Analysis of Time-series data," Third Edition, Wiley 2014

Faculty HOD 11/2 Dean Academics 11/2/26

UNIT-4

Introduction to Least Squares Estimation in Linear Regression Models

Lecture-1

Least Squares Estimation is a fundamental method used in statistics and machine learning to estimate the unknown parameters of a linear regression model. It provides a systematic way to determine the best-fitting line or function that explains the relationship between a dependent variable and one or more independent variables. The main objective of least squares is to minimize the difference between the observed values and the predicted values produced by the regression model.

Linear Regression Model

A linear regression model expresses the relationship between variables using a linear equation. In the simplest case, where only one predictor variable is involved, the model can be written as:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

Where:

- Y_i= dependent (response) variable
- X_i= independent (predictor) variable
- β_0= intercept
- β_1= slope coefficient
- ϵ_i= random error term

The goal is to estimate unknown parameters $\beta_0$and β_1.

When multiple predictors are involved, the model extends to multiple linear regression, where several independent variables jointly explain the variation in the dependent variable.

When more than one predictor is used:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon$$

Concept of Least Squares Estimation

In real datasets, observed points rarely lie perfectly on a straight line due to variability and measurement errors. Least squares estimation addresses this by choosing parameter values that minimize the total squared difference between observed values and predicted values. These differences are called residuals.

These differences are called **residuals**:

$$e_i = Y_i - \hat{Y}_i$$

Where:

- Y_i= actual value
- $\hat{Y}_i$= predicted value

A residual represents the vertical distance between the actual observation and the regression line. Instead of minimizing the sum of residuals directly, the method minimizes the sum of squared residuals, ensuring that positive and negative errors do not cancel each other out and that larger errors receive greater penalty.

The quantity minimized is known as the Sum of Squared Errors:

$$SSE = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2$$

Where:

- Y_i= actual value
- $\hat{Y}_i$= predicted value

This minimization leads to estimates of the regression coefficients that produce the best possible fit according to the least squares criterion.

Derivation of Least Squares Estimates

To find the optimal regression line, the squared error expression is written in terms of unknown parameters. By taking partial derivatives with respect to the intercept and slope and setting them equal to zero, a system of equations known as the normal equations is obtained. Solving these equations yields formulas for the estimated slope and intercept.

The estimated slope measures the strength and direction of the linear relationship between variables, while the intercept adjusts the position of the regression line so that the overall error is minimized.

Predicted value:

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$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

Objective function:

$$S = \sum (Y_i - \beta_0 - \beta_1 X_i)^2$$

To minimize S, take partial derivatives and set to zero.

Normal Equations

$$\sum Y_i = n\beta_0 + \beta_1 \sum X_i$$
$$\sum X_i Y_i = \beta_0 \sum X_i + \beta_1 \sum X_i^2$$

Estimated Coefficients

Slope:

$$\hat{\beta}_1 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

Intercept:

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

These formulas give the **best-fit regression line**.

Matrix Representation of Least Squares

For datasets with multiple predictors, matrix notation provides a compact and powerful way to express linear regression. The model can be written as:

$$Y = X\beta + \epsilon$$

Where:

- Y= response vector
- X= design matrix
- β= parameter vector
- ϵ= error vector

where $Y$is a vector of observed responses, $X$is the design matrix containing predictor values, $\beta$is the vector of unknown coefficients, and ϵrepresents the error vector.

Using matrix algebra, the least squares solution becomes:

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

This formulation is widely used in modern data science and machine learning because it simplifies computations and allows efficient implementation using linear algebra techniques.

Geometric Interpretation

Least squares estimation also has a geometric meaning. The observed data vector can be viewed as a point in a high-dimensional space, and the regression model finds the projection of this point onto the space spanned by the predictor variables. The residual vector is perpendicular to the space defined by the predictors, which ensures that the chosen regression line is the closest possible approximation to the data.

This means:

$$\boldsymbol{X^T(Y - X\hat{\beta}) = 0}$$

Assumptions of Least Squares Estimation

For least squares estimates to have desirable statistical properties, several assumptions are generally made. The relationship between predictors and response is assumed to be linear, observations are considered independent, and the variance of errors remains constant across all observations. In multiple regression, predictors should not be perfectly correlated with each other, and the error terms are often assumed to follow a normal distribution when hypothesis testing or confidence intervals are required.

Properties of Least Squares Estimators

Least squares estimators possess important theoretical properties. They are unbiased, meaning that their expected value equals the true parameter value. Under the classical assumptions, they also have minimum variance among all linear unbiased estimators, a result known as the Gauss–Markov theorem. As the sample size increases, the estimates become more accurate and stable.

Interpretation of Regression Coefficients

The intercept represents the expected value of the dependent variable when all predictors are zero. The slope coefficients describe how the dependent variable changes with respect to each independent variable. For example, in a model predicting marks based on study hours, a slope of two would indicate that each additional hour of study increases the expected marks by two units.

Advantages of Least Squares Estimation

Least squares estimation is simple to understand and computationally efficient. It provides a closed-form mathematical solution and forms the basis of many advanced statistical and machine learning techniques. Because of its strong theoretical foundation, it is widely applied in fields such as economics, engineering, and image processing.

Limitations of Least Squares

Despite its usefulness, least squares estimation has certain drawbacks. It is sensitive to outliers because squaring the errors amplifies the effect of extreme values. It also assumes a linear relationship between variables and may perform poorly when predictors are highly correlated or when model assumptions are violated.

Applications

Least squares estimation is used extensively in predictive modeling, financial forecasting, medical research, signal processing, and computer vision. Many modern algorithms, including neural networks and optimization methods, are conceptually related to minimizing squared error.

Conclusion

Least Squares Estimation plays a central role in linear regression analysis by providing a reliable method for estimating model parameters. By minimizing the sum of squared residuals, it identifies the line that best represents the relationship between variables. Its mathematical simplicity, strong theoretical properties, and wide applicability make it one of the most important tools in statistics, research methodology, and data science.

References/Text Books

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Statistical Inference in Linear Regression and Prediction of New Observations

Lecture-2

Statistical inference in linear regression focuses on drawing conclusions about the population parameters based on sample data. After estimating regression coefficients using least squares, the next important step is to evaluate the reliability of these estimates, test hypotheses about relationships between variables, and make predictions for new observations. Inference allows researchers to determine whether observed patterns are statistically significant or simply due to random variation.

Linear Regression Model Framework

Consider the simple linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

where Y_i is the response variable, X_i is the predictor, β_0 and β_1 are unknown parameters, and ϵ_i represents the random error term. The error terms are typically assumed to be independently and normally distributed with mean zero and variance σ^2.

The estimated regression line obtained from least squares is:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

Statistical inference examines how close the estimated parameters $\hat{\beta}_0$ and $\hat{\beta}_1$ are to the true parameters.

Sampling Distribution of Regression Coefficients

The least squares estimators are random variables because they depend on sample data. Under classical assumptions:

$$E(\hat{\beta}_1) = \beta_1, E(\hat{\beta}_0) = \beta_0$$

This shows that the estimators are unbiased. Their variances are:

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{\sum (X_i - \bar{X})^2}$$
$$Var(\hat{\beta}_0) = \sigma^2 \left[\frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2}\right]$$

Since $\sigma^2$is usually unknown, it is estimated using the residual mean square:

$$s^2 = \frac{\sum(Y_i - \hat{Y}_i)^2}{n-2}$$

Hypothesis Testing for Regression Coefficients

Statistical inference often involves testing whether a predictor variable significantly influences the response.

A common hypothesis test for the slope is:

$$H_0: \beta_1 = 0 \text{(no relationship)}$$
$$H_1: \beta_1 \neq 0$$

The test statistic is:

$$t = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)}$$

where

$$SE(\hat{\beta}_1) = \sqrt{\frac{s^2}{\sum(X_i - \bar{X})^2}}$$

If the calculated t-value exceeds the critical value from the t-distribution with $n - 2$degrees of freedom, the null hypothesis is rejected, indicating that the predictor significantly affects the response.

Example:
Suppose a regression model estimates:

$$\hat{Y} = 40 + 3X$$

If the standard error of $\hat{\beta}_1$is 0.8, then:

$$t = \frac{3}{0.8} = 3.75$$

A large t-value suggests that study hours significantly influence marks.

Confidence Intervals for Regression Parameters

Instead of a single estimate, inference often provides a range of plausible values for parameters.

A confidence interval for the slope is:

$$\hat{\beta}_1 \pm t_{\alpha/2,n-2}\, SE(\hat{\beta}_1)$$

Similarly, a confidence interval can be constructed for the intercept.

Example:
If $\hat{\beta}_1 = 3$, $SE = 0.8$, and the critical value is 2.0, then:

$$3 \pm 2(0.8) = (1.4,4.6)$$

This means the true slope likely lies between 1.4 and 4.6.

Analysis of Variance (ANOVA) in Regression

ANOVA decomposes total variability into explained and unexplained parts:

$$SST = SSR + SSE$$

where:

- SST= total sum of squares
- SSR= regression sum of squares
- SSE= error sum of squares

The coefficient of determination is:

$$R^2 = \frac{SSR}{SST}$$

This measures the proportion of variability in the response explained by the predictors.

Prediction of New Observations

One of the main uses of linear regression is predicting future or unseen data points.

The predicted mean response at a new value $X_0$is:

$$\hat{Y}_0 = \hat{\beta}_0 + \hat{\beta}_1 X_0$$

However, predictions contain uncertainty, which is measured using intervals.

Confidence Interval for Mean Response

This interval estimates the average value of Yat a specific X_0:

$$\hat{Y}_0 \pm t_{\alpha/2,n-2}\, s \sqrt{\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{\sum(X_i - \bar{X})^2}}$$

This interval is relatively narrow because it refers to the mean response.

Prediction Interval for a New Observation

When predicting a single future value, additional variability must be considered. The prediction interval is:

$$\hat{Y}_0 \pm t_{\alpha/2,n-2}\, s \sqrt{1 + \frac{1}{n} + \frac{(X_0 - \bar{X})^2}{\sum(X_i - \bar{X})^2}}$$

Prediction intervals are wider than confidence intervals because they account for both model uncertainty and random error.

Example:
Suppose the regression equation is:

$$\hat{Y} = 40 + 3X$$

If a student studies $X_0 = 5$hours:

$$\hat{Y}_0 = 40 + 3(5) = 55$$

A confidence interval might be (52, 58), representing the average marks of students studying five hours, while a prediction interval might be (45, 65), representing the marks of an individual student.

Interpretation and Practical Importance

Statistical inference helps determine whether relationships observed in the sample are meaningful for the broader population. Hypothesis tests reveal the significance of predictors, confidence intervals quantify estimation uncertainty, and prediction intervals allow forecasting of future observations. Together, these tools transform regression from a simple curve-fitting method into a powerful framework for decision-making and scientific analysis.

Conclusion

Statistical inference in linear regression provides the theoretical foundation for understanding parameter estimates, testing relationships between variables, and quantifying uncertainty. By using hypothesis testing, confidence intervals, and ANOVA, researchers can evaluate the reliability of regression models. Prediction techniques further extend regression analysis by enabling estimation of future outcomes along with measures of uncertainty, making linear regression an essential tool in statistics, research methodology, and applied data science.

References/Text Books

1 " Introduction To Time Series Analysis And Forecasting, 2nd Edition, Wiley Series In Probability And Statistics, By Douglas C. Montgomery, Cheryl L. Jen(2015)"

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Model Adequacy Checking

Lecture-3

Model adequacy checking in linear regression refers to the process of evaluating whether a fitted regression model appropriately represents the data and satisfies the underlying statistical assumptions. Even if regression coefficients are estimated correctly, the model may still be unsuitable if assumptions such as linearity, independence, constant variance, and normality of errors are violated. Therefore, model adequacy checking ensures that statistical inference and predictions made from the model are reliable.

Purpose of Model Adequacy Checking

After fitting a regression model, it is necessary to verify whether the model captures the true relationship between variables. Adequacy checking helps in:

- Identifying violations of model assumptions
- Detecting outliers or influential observations
- Evaluating goodness of fit
- Improving model performance through transformation or variable selection

A model is considered adequate if the residuals behave like random noise with no systematic pattern.

Residual Analysis

Residuals are defined as the differences between observed and predicted values:

$$e_i = Y_i - \hat{Y}_i$$

Residual analysis is the most important tool for checking model adequacy because residuals contain information about model errors.

If the model is appropriate, residuals should:

- Have a mean close to zero
- Show no pattern when plotted against predictors
- Have constant variance
- Follow an approximately normal distribution

Residual Plots

A plot of residuals versus fitted values is commonly used to examine the linearity and variance assumptions. Ideally, points should appear randomly scattered around zero.

Interpretation of patterns:

- A curved pattern suggests a non-linear relationship, indicating that the model may require transformation or additional predictors.
- A funnel-shaped pattern indicates heteroscedasticity, meaning the variance of errors is not constant.
- Clusters or systematic structures imply missing variables or incorrect model form.

Example:
Suppose marks are predicted using study hours. If residuals increase as study hours increase, the model may not capture variability properly, and a transformation such as logarithmic scaling may be needed.

Normal Probability Plot of Residuals

Many inferential procedures assume normally distributed errors. A normal probability plot (Q–Q plot) compares ordered residuals with theoretical normal values.

If residuals lie approximately along a straight line, the normality assumption is reasonable. Large deviations from the line suggest skewness or heavy tails.

Checking Independence of Errors

Independence is especially important in time-series or sequential data. Correlated errors can lead to incorrect statistical conclusions.

A common method is examining residuals plotted against time or observation order. If patterns or trends appear, autocorrelation may exist.

The Durbin–Watson statistic is often used to test for autocorrelation:

$$DW = \frac{\sum_{i=2}^{n}(e_i - e_{i-1})^2}{\sum_{i=1}^{n} e_i^2}$$

Values near 2 indicate independence, while values close to 0 or 4 suggest positive or negative autocorrelation.

Detection of Outliers and Influential Observations

Outliers are observations with unusually large residuals. Influential observations significantly affect the estimated regression line.

Standardized residuals help identify extreme observations:

$$r_i = \frac{e_i}{s\sqrt{1 - h_{ii}}}$$

where h_{ii} represents leverage values from the hat matrix.

Cook's distance is another measure used to assess influence:

$$D_i = \frac{e_i^2}{ps^2} \cdot \frac{h_{ii}}{(1-h_{ii})^2}$$

Large values of Cook's distance indicate observations that strongly impact the model.

Example:
If one student studies very few hours but scores extremely high marks, that observation may distort the regression line and should be examined carefully.

Goodness-of-Fit Measures

Model adequacy is also evaluated using statistical metrics.

The coefficient of determination is:

$$R^2 = \frac{SSR}{SST}$$

where SSR is regression sum of squares and SST is total sum of squares. A higher R^2 indicates that the model explains a larger proportion of variability in the response variable.

Adjusted R^2 is often preferred in multiple regression because it accounts for the number of predictors.

Lack-of-Fit Testing

In some cases, formal tests can be used to determine whether the chosen model form is appropriate. A lack-of-fit test compares the variability explained by the model with the variability due to pure error. A significant lack-of-fit indicates that the model structure may be incorrect, suggesting the need for additional variables or nonlinear terms.

Remedies for an Inadequate Model

If adequacy checks reveal problems, several corrective actions can be taken:

- Transforming variables using logarithmic or square-root transformations

- Adding interaction terms or polynomial terms
- Removing outliers or influential points after investigation
- Using weighted least squares when variance is not constant
- Including missing predictors to improve model accuracy

Illustrative Example

Consider a regression model predicting crop yield based on fertilizer amount:

$$\hat{Y} = 20 + 4X$$

If a residual plot shows a curved pattern, it suggests that yield increases rapidly at first but slows later. A quadratic model such as

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2$$

may provide a better fit.

Conclusion

Model adequacy checking is a critical step in regression analysis that ensures the validity of statistical inference and predictions. By analyzing residuals, checking assumptions, identifying outliers, and evaluating goodness-of-fit measures, researchers can determine whether a regression model accurately represents the data. Adequate models lead to reliable conclusions, while inadequate models require refinement through transformations, additional predictors, or alternative modeling approaches.

References/Text Books

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2 "Master Time Series Data Processing, Visualization, And Modeling Using Python Dr. Avishek Pal Dr. Pks Prakash (2017)"

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Variable Selection Methods in Regression

Lecture-4

Variable selection in regression refers to the process of identifying the most relevant predictor variables that should be included in a regression model. In many real-world problems, datasets may contain a large number of potential explanatory variables, but not all of them contribute significantly to predicting the response variable. Including unnecessary variables can make the model complex, increase variance, reduce interpretability, and sometimes lead to overfitting. Therefore, selecting an appropriate subset of variables is essential for building efficient and reliable regression models.

Need for Variable Selection

When too many predictors are included in a regression model, several problems may arise. Multicollinearity can occur when predictors are highly correlated, leading to unstable coefficient estimates. Models may become overly complex and difficult to interpret. Additionally, unnecessary variables increase computational cost and may reduce prediction accuracy on new data. Variable selection helps in improving model simplicity, interpretability, and generalization performance.

Best Subset Selection

Best subset selection involves fitting regression models for all possible combinations of predictor variables and choosing the best model according to a specific criterion. If there are $p$predictors, then 2^ppossible models exist. Each model is evaluated using measures such as adjusted R^2, Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), or residual sum of squares.

The general regression model considered is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon$$

Best subset selection guarantees that the chosen model is optimal under the selected criterion, but it becomes computationally expensive when the number of predictors is large.

Example:
Suppose a model predicts student performance using variables such as study hours, attendance, sleep time, and internet usage. Best subset selection evaluates all combinations of these variables and identifies the subset that provides the best predictive performance.

Forward Selection Method

Forward selection begins with an empty model containing only the intercept. Variables are added one at a time based on their statistical significance or contribution to model improvement. At each step, the variable that produces the greatest reduction in error or improvement in a chosen criterion is included.

The procedure continues until adding additional variables no longer significantly improves the model.

Example:
A regression model predicting crop yield may start with no predictors, then add fertilizer amount, followed by rainfall, and finally temperature if each step improves model accuracy.

Backward Elimination Method

Backward elimination starts with the full model containing all candidate predictors. Variables are removed sequentially based on their statistical insignificance. Typically, the predictor with the highest p-value greater than a specified significance level is removed at each step.

This process continues until all remaining predictors are statistically significant.

Example:
A health dataset might include age, weight, exercise time, diet score, and screen time. If screen time has a very high p-value, it may be removed first, followed by other insignificant variables.

Stepwise Selection Method

Stepwise selection combines forward selection and backward elimination. Variables can be added or removed at different stages of the model-building process. After adding a new predictor, the method checks whether previously included variables have become insignificant and removes them if necessary.

This approach provides flexibility and often results in a more balanced model compared to purely forward or backward methods.

Selection Criteria for Comparing Models

Various statistical criteria are used to evaluate and compare regression models during variable selection.

The adjusted coefficient of determination is:

$$R^2_{adj} = 1 - \frac{SSE/(n-p-1)}{SST/(n-1)}$$

Adjusted R^2penalizes the inclusion of unnecessary variables, unlike ordinary R^2, which always increases when new predictors are added.

The Akaike Information Criterion is defined as:

$$AIC = n\ln\left(\frac{SSE}{n}\right) + 2p$$

The Bayesian Information Criterion is:

$$BIC = n\ln\left(\frac{SSE}{n}\right) + p\ln(n)$$

Lower values of AIC or BIC indicate a better balance between model fit and complexity.

Regularization-Based Variable Selection

Modern regression techniques often perform variable selection through regularization, where a penalty is added to the least squares objective function.

In ridge regression, the objective function is:

$$\sum(Y_i - \hat{Y}_i)^2 + \lambda\sum\beta_j^2$$

This method reduces coefficient magnitudes but does not set them exactly to zero.

In LASSO regression, the penalty uses absolute values:

$$\sum(Y_i - \hat{Y}_i)^2 + \lambda\sum \mid \beta_j \mid$$

LASSO can shrink some coefficients exactly to zero, effectively performing automatic variable selection.

Elastic Net combines both ridge and LASSO penalties, balancing stability and sparsity.

Multicollinearity Considerations

Variable selection must also account for multicollinearity, which occurs when predictors are highly correlated. High multicollinearity inflates the variance of coefficient estimates and makes interpretation difficult. Variance Inflation Factor (VIF) is often used to detect this issue:

$$VIF_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is obtained by regressing the j-th predictor on all other predictors. Large VIF values indicate potential multicollinearity problems.

Advantages of Variable Selection

Variable selection improves interpretability by reducing unnecessary predictors and helps avoid overfitting. It can enhance prediction accuracy, reduce model complexity, and make statistical inference more reliable.

Limitations

Despite its benefits, variable selection has challenges. Different selection methods may lead to different models, and stepwise approaches may sometimes select variables due to random variation rather than true importance. Overreliance on automated selection procedures without domain knowledge can also lead to misleading results.

References/Text Books

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Generalized and Weighted Least Squares

Lecture-5

In classical linear regression, the ordinary least squares (OLS) method assumes that the error terms have constant variance and are uncorrelated. However, in many practical situations these assumptions are violated. When the variance of errors is not constant (heteroscedasticity) or when errors are correlated, more advanced estimation techniques such as **Weighted Least Squares (WLS)** and **Generalized Least Squares (GLS)** are used. These methods extend the least squares framework to produce efficient and unbiased parameter estimates under more general conditions.

Limitations of Ordinary Least Squares

Consider the linear regression model in matrix form:

$$Y = X\beta + \epsilon$$

where $Y$is the response vector, $X$is the design matrix, $\beta$is the vector of unknown parameters, and $\epsilon$is the error vector.

Ordinary least squares assumes:

$$E(\epsilon) = 0, Var(\epsilon) = \sigma^2 I$$

This means that all error terms have equal variance and are uncorrelated. When these assumptions fail, OLS estimates remain unbiased but are no longer efficient, and standard errors may be incorrect. This motivates the use of weighted and generalized least squares.

Weighted Least Squares (WLS)

Weighted least squares is used when error variances are unequal but still uncorrelated. Suppose the variance of each observation is different:

$$Var(\epsilon_i) = \sigma_i^2$$

Instead of minimizing the ordinary sum of squared residuals, WLS minimizes a weighted sum:

$$S = \sum_{i=1}^{n} w_i \, (Y_i - \hat{Y}_i)^2$$

where $w_i = \frac{1}{\sigma_i^2}$is the weight assigned to the i-th observation. Observations with smaller variance receive larger weights because they contain more reliable information.

In matrix form, define a diagonal weight matrix:

$$W = \begin{bmatrix} w_1 & 0 & \cdots & 0 \\ 0 & w_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & w_n \end{bmatrix}$$

The weighted least squares estimator becomes:

$$\hat{\beta}_{WLS} = (X^T W X)^{-1} X^T W Y$$

Interpretation of Weighted Least Squares

Weighted least squares adjusts the influence of each observation on the regression line. Data points with high variability contribute less to parameter estimation, while more precise observations have greater impact.

Example:
Suppose crop yield measurements are taken using two instruments, where one instrument produces more variable readings. Assigning lower weights to measurements from the less accurate instrument improves the reliability of the regression model.

Transformation Approach to WLS

Weighted least squares can be interpreted as transforming the data. If each observation is divided by the square root of its variance, the transformed model satisfies the assumptions of ordinary least squares. This transformation converts heteroscedastic data into homoscedastic data, allowing standard regression techniques to be applied.

Generalized Least Squares (GLS)

Generalized least squares extends WLS by handling both unequal variances and correlated errors. Suppose the covariance matrix of the error vector is:

$$Var(\epsilon) = \Sigma$$

where $\Sigma$is a non-diagonal matrix. The GLS method minimizes the quadratic form:

$$(Y - X\beta)^T \Sigma^{-1} (Y - X\beta)$$

The GLS estimator is:

$$\hat{\beta}_{GLS} = (X^T\Sigma^{-1}X)^{-1}X^T\Sigma^{-1}Y$$

This estimator reduces to the ordinary least squares estimator when $\Sigma = \sigma^2 I$, and to weighted least squares when $\Sigma$is diagonal.

Geometric and Statistical Interpretation

Generalized least squares can be viewed as transforming the original data using a matrix that removes correlation and stabilizes variance. After transformation, the regression problem behaves like a standard least squares problem. This transformation ensures that the residuals become independent with equal variance, restoring the efficiency of parameter estimates.

Example of GLS Application

Consider a time-series regression where consecutive observations are correlated. For example, daily temperature readings often show autocorrelation because today's temperature is related to yesterday's temperature. Using ordinary least squares in such a case leads to inefficient estimates. GLS accounts for the correlation structure by incorporating the covariance matrix Σ, producing more accurate coefficient estimates and valid statistical inference.

Comparison between OLS, WLS, and GLS

Ordinary least squares treats all observations equally and assumes constant variance and independence. Weighted least squares addresses unequal variances by assigning weights to observations. Generalized least squares provides the most general framework by allowing both heteroscedasticity and correlation among errors.

References/Text Books

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Regression Models for General Time Series Data

Lecture-6

Regression models for general time series data are used to analyze relationships between variables when observations are collected sequentially over time. Unlike standard regression, time series data often exhibit characteristics such as trend, seasonality, autocorrelation, and time-dependent variability. These properties violate the classical assumptions of independent errors, requiring specialized regression approaches that incorporate temporal structure into the model.

Time series regression combines traditional regression analysis with stochastic processes to model how a response variable evolves over time while being influenced by predictors and past values.

Basic Time Series Regression Model

A general time series regression model can be written as:

$$Y_t = \beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t} + \cdots + \beta_p X_{pt} + \epsilon_t$$

where Y_t is the response at time t, X_{jt} represents predictors at time t, and ϵ_t denotes the error term. Unlike classical regression, the error terms may be correlated:

$$Cov(\epsilon_t, \epsilon_{t-k}) \neq 0$$

This autocorrelation is a defining feature of time series data.

Trend Models in Time Series Regression

Many time series exhibit a long-term increase or decrease called a trend. A regression model including a time index can capture this behavior:

$$Y_t = \beta_0 + \beta_1 t + \epsilon_t$$

A positive β_1 indicates an increasing trend over time.

Example:
If annual sales data show steady growth, the model

$$Y_t = 100 + 5t$$

implies that sales increase by 5 units each year.

Nonlinear trends can be modeled using polynomial regression:

$$Y_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \epsilon_t$$

Seasonal Regression Models

Seasonal patterns occur when data repeat periodically, such as monthly or quarterly effects. These patterns can be incorporated using indicator (dummy) variables or trigonometric terms.

A seasonal regression with dummy variables is:

$$Y_t = \beta_0 + \beta_1 D_{1t} + \beta_2 D_{2t} + \cdots + \epsilon_t$$

where D_{jt}represents seasonal indicators.

Alternatively, sinusoidal models capture periodic behavior:

$$Y_t = \beta_0 + \alpha \sin(\omega t) + \gamma \cos(\omega t) + \epsilon_t$$

Example:
Electricity demand often increases in summer and winter. Seasonal terms help explain these repeating patterns.

Autoregressive Error Models

In many time series, residuals are correlated across time. A common approach is to model the errors as an autoregressive process.

An autoregressive model of order one, AR(1), is:

$$\epsilon_t = \rho \epsilon_{t-1} + u_t$$

where $|\rho| < 1$and $u_t$is white noise.

The regression model becomes:

$$Y_t = X_t \beta + \epsilon_t$$

with correlated errors. Estimation is often performed using generalized least squares to account for autocorrelation.

Dynamic Regression Models

Dynamic regression includes lagged values of the response or predictors:

$$Y_t = \beta_0 + \beta_1 X_t + \phi Y_{t-1} + \epsilon_t$$

Such models capture delayed effects.

Example:
Today's temperature may depend on today's humidity and yesterday's temperature. Including Y_{t-1} allows the model to incorporate memory of past observations.

Distributed Lag Models

In some situations, a predictor affects the response over several future time periods. Distributed lag models include multiple lagged predictors:

$$Y_t = \beta_0 + \beta_1 X_t + \beta_2 X_{t-1} + \beta_3 X_{t-2} + \epsilon_t$$

This is useful in economics, where policy changes may influence outcomes gradually.

Regression with ARIMA Errors

A powerful framework combines regression with ARIMA (AutoRegressive Integrated Moving Average) processes for errors:

$$Y_t = X_t \beta + n_t$$

where n_t follows an ARIMA process. This approach separates deterministic effects (predictors) from stochastic temporal dynamics.

For example, an ARMA(1,1) error structure is:

$$n_t = \phi n_{t-1} + a_t + \theta a_{t-1}$$

This method is widely used in forecasting because it models both explanatory variables and time dependence.

Estimation Techniques

Because time series data often violate independence assumptions, ordinary least squares may produce inefficient estimates. Techniques commonly used include:

- Generalized least squares to handle correlated errors
- Maximum likelihood estimation for ARIMA-based regression
- Iterative procedures to estimate regression coefficients and autocorrelation parameters simultaneously

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Model Adequacy for Time Series Regression

Adequacy checking focuses on residual autocorrelation. After fitting a model, residuals should resemble white noise.

The sample autocorrelation function (ACF) is used to examine residual dependence:

$$r_k = \frac{\sum_{t=k+1}^{n}(e_t - \bar{e})(e_{t-k} - \bar{e})}{\sum_{t=1}^{n}(e_t - \bar{e})^2}$$

If significant autocorrelation remains, the model may require additional lag terms or a different error structure.

Example of a Time Series Regression Model

Suppose monthly sales depend on advertising expenditure and a time trend:

$$Y_t = 50 + 3X_t + 2t + \epsilon_t$$

Here, advertising explains short-term variation, while the time variable captures long-term growth. If residuals show autocorrelation, an AR(1) error term may be added:

$$\epsilon_t = 0.6\epsilon_{t-1} + u_t$$

This results in a more realistic model that accounts for temporal dependence.

References/Text Books

1 " Introduction To Time Series Analysis And Forecasting, 2nd Edition, Wiley Series In Probability And Statistics, By Douglas C. Montgomery, Cheryl L. Jen(2015)"

2 "Master Time Series Data Processing, Visualization, And Modeling Using Python Dr. Avishek Pal Dr. Pks Prakash (2017)"

3 Ruey S. Tsay "Analysis of Time-series data," Third Edition, Wiley 2014

Exponential Smoothing

Lecture-7

Exponential smoothing is a widely used forecasting technique for time series data that assigns exponentially decreasing weights to past observations. Unlike simple moving averages, which give equal importance to recent values within a window, exponential smoothing places more emphasis on recent data while still retaining information from older observations. Because of its simplicity, efficiency, and strong forecasting performance, exponential smoothing is commonly applied in business forecasting, economics, engineering, and demand prediction.

Basic Idea of Exponential Smoothing

The main objective of exponential smoothing is to generate forecasts by updating previous predictions using new observations. The method assumes that recent observations provide more relevant information about future values than older data.

Let Y_t denote the observed value at time t, and let F_t denote the forecast for period t. The smoothing parameter α controls how much weight is given to recent observations, where $0 < \alpha < 1$.

Simple Exponential Smoothing

Simple exponential smoothing is suitable for time series that do not contain strong trend or seasonal patterns. The forecast is updated using the formula:

$$F_{t+1} = \alpha Y_t + (1 - \alpha) F_t$$

This equation shows that the new forecast is a weighted average of the current observation and the previous forecast. A larger value of α makes the forecast respond quickly to recent changes, while a smaller value produces smoother forecasts.

An alternative form of the updating equation is:

$$F_{t+1} = F_t + \alpha (Y_t - F_t)$$

where $(Y_t - F_t)$ is the forecast error at time t.

Example:

Suppose the smoothing parameter is $\alpha = 0.3$, the previous forecast is $F_t = 100$, and the observed value is $Y_t = 110$. The updated forecast becomes:

$$F_{t+1} = 100 + 0.3(110 - 100) = 103$$

The forecast moves closer to the observed value but does not change drastically.

Weighted Average Interpretation

Exponential smoothing can be expanded recursively:

$$F_{t+1} = \alpha Y_t + \alpha(1-\alpha)Y_{t-1} + \alpha(1-\alpha)^2 Y_{t-2} + \cdots$$

This shows that older observations receive exponentially smaller weights. The method therefore avoids storing large historical datasets, making it computationally efficient.

Holt's Linear Trend Method (Double Exponential Smoothing)

When a time series contains a trend, simple exponential smoothing is not sufficient. Holt's method extends the model by including separate equations for the level and trend components.

Level equation:

$$L_t = \alpha Y_t + (1-\alpha)(L_{t-1} + T_{t-1})$$

Trend equation:

$$T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1}$$

Forecast equation:

$$F_{t+h} = L_t + hT_t$$

Here, $L_t$represents the smoothed level, $T_t$represents the trend, and $\beta$is the trend smoothing parameter.

Example:
If sales increase steadily each month, Holt's method captures both the current level of sales and the upward trend to produce more accurate forecasts.

Holt–Winters Seasonal Exponential Smoothing

Many real-world time series exhibit seasonal patterns, such as weekly or yearly cycles. The Holt–Winters method incorporates level, trend, and seasonality.

For an additive seasonal model:

Level:

$$L_t = \alpha(Y_t - S_{t-m}) + (1-\alpha)(L_{t-1} + T_{t-1})$$

Trend:

$$T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1}$$

Seasonal component:

$$S_t = \gamma(Y_t - L_t) + (1-\gamma)S_{t-m}$$

Forecast:

$$F_{t+h} = L_t + hT_t + S_{t-m+h}$$

Here, $m$is the seasonal period, and $\gamma$is the seasonal smoothing parameter.

Example:
Monthly electricity demand often shows yearly seasonal patterns. The Holt–Winters method captures both the growth trend and seasonal fluctuations.

Choice of Smoothing Parameters

The parameters α, β, and γdetermine how quickly the model adapts to changes. These parameters are typically chosen by minimizing forecast error measures such as Mean Squared Error (MSE):

$$MSE = \frac{1}{n}\sum(Y_t - F_t)^2$$

Higher values of smoothing parameters make the model more responsive but may increase noise sensitivity.

References/Text Books

1 " Introduction To Time Series Analysis And Forecasting, 2nd Edition, Wiley Series In Probability And Statistics, By Douglas C. Montgomery, Cheryl L. Jen(2015)"

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First Order and Second Order Exponential Smoothing

Lecture-8

Exponential smoothing techniques are commonly classified into first order and second order methods depending on how many components of the time series they model. First order exponential smoothing is used for data without trend, while second order exponential smoothing extends the idea to capture trend behavior. These methods provide simple yet powerful tools for forecasting time series data.

First Order Exponential Smoothing

First order exponential smoothing, also called **single exponential smoothing**, is suitable for time series that fluctuate around a constant mean and do not show a clear upward or downward trend. The idea is to update forecasts by combining the latest observation with the previous forecast.

Let Y_t be the observed value at time t, and F_t be the forecast at time t. The updating equation is:

$$F_{t+1} = \alpha Y_t + (1 - \alpha) F_t$$

where $0 < \alpha < 1$ is the smoothing constant.

Another useful form of the equation is:

$$F_{t+1} = F_t + \alpha (Y_t - F_t)$$

The term $(Y_t - F_t)$ represents the forecast error. The parameter α determines how quickly the forecast reacts to new data. A large value of α gives more weight to recent observations, while a small value produces smoother forecasts.

Example:
Suppose daily demand has no trend. Let $F_t = 200$, $Y_t = 220$, and $\alpha = 0.4$. Then

$$F_{t+1} = 200 + 0.4(220 - 200) = 208$$

The forecast moves toward the new observation but remains stable.

Characteristics

- Assumes no trend or seasonality
- Simple to compute

- Useful for short-term forecasting

Second Order Exponential Smoothing

Second order exponential smoothing, also known as **double exponential smoothing** or Holt’s linear method, is designed for time series that exhibit a trend. Instead of using only one smoothing equation, it introduces two components: the level and the trend.

The level component is updated as:

$$L_t = \alpha Y_t + (1-\alpha)(L_{t-1} + T_{t-1})$$

The trend component is updated as:

$$T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1}$$

The forecast for $h$steps ahead is:

$$F_{t+h} = L_t + hT_t$$

where:

- $L_t$is the smoothed level at time t
- $T_t$is the estimated trend
- $\alpha$is the level smoothing parameter
- $\beta$is the trend smoothing parameter

Example:
Consider monthly sales increasing steadily. Suppose

$$L_{t-1} = 500, T_{t-1} = 20, Y_t = 540, \alpha = 0.3, \beta = 0.2$$

First compute the new level:

$$L_t = 0.3(540) + 0.7(500 + 20) = 162 + 364 = 526$$

Then update the trend:

$$T_t = 0.2(526 - 500) + 0.8(20) = 5.2 + 16 = 21.2$$

The next forecast is:

$$F_{t+1} = 526 + 21.2 = 547.2$$

This method captures both the current level and the upward trend in sales.

Key Differences Between First Order and Second Order Methods

First order exponential smoothing uses a single equation and assumes the time series has no trend. It produces stable forecasts when data fluctuate around a constant mean. Second order exponential smoothing introduces an additional trend component, allowing forecasts to increase or decrease over time.

First order smoothing requires only one parameter α, while second order smoothing requires two parameters $\alpha$and β. Because of the trend adjustment, second order methods provide better forecasts for growing or declining time series.

References/Text Books

1 " Introduction To Time Series Analysis And Forecasting, 2nd Edition, Wiley Series In Probability And Statistics, By Douglas C. Montgomery, Cheryl L. Jen(2015)"

2 "Master Time Series Data Processing, Visualization, And Modeling Using Python Dr. Avishek Pal Dr. Pks Prakash (2017)"

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